# CVHS ENGINEERING

Iso + ortho sketch sheet | 5 mm grid

Third-angle layout. Sheet v0.3, Nov 2026.

## Example A: the L piece

Worked example. One cube = two squares.
Each colour is one side of the block. Find it in the picture.

Name _____

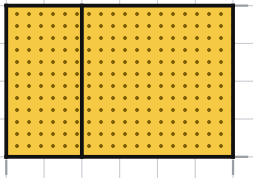
Date _____ Period _____

Colour = the side you are looking at

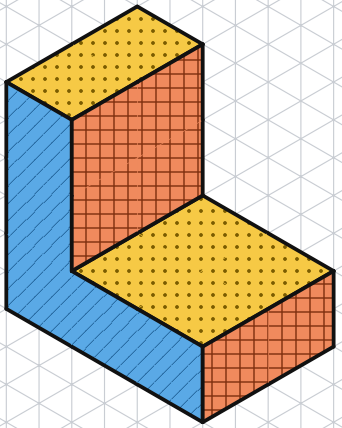
TOP FRONT RIGHT

### TOP VIEW

Look straight down.
You only see the top faces: a 3 by 2 box with a line where the step drops.



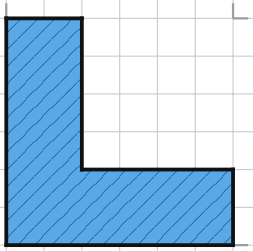
### ISOMETRIC




The picture. Three sides, three colours.
Grey dashed lines show how the views line up with each other.

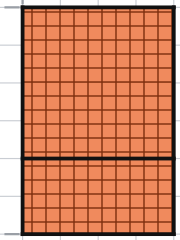
### FRONT VIEW

Look straight at it. You see the L.
Width goes across, height goes up.



### RIGHT SIDE VIEW

Look from the right side.
Depth goes across, height goes up.
The front of the block is on the LEFT edge of this view.



Views line up: TOP sits straight above FRONT, RIGHT sits straight beside it. Leave two squares between views. One cube = two squares.

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## Example B: three stairs

Worked example. One cube = two squares.
Each colour is one side of the block. Find it in the picture.

Name _____

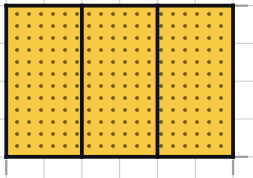
Date _____ Period _____

Colour = the side you are looking at

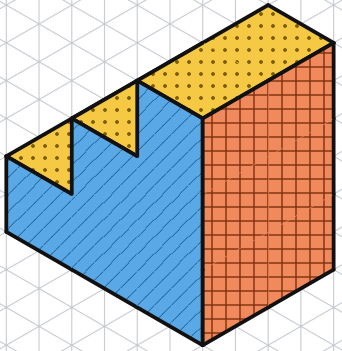
TOP FRONT RIGHT

### TOP VIEW

Three steps = three bands.
Every change in height makes a line.



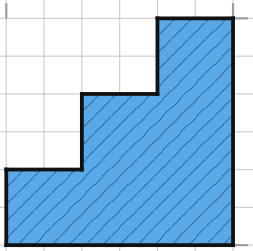
### ISOMETRIC




Count cubes: 1, 2, 3 tall, 2 deep.
Solid line = an edge you can see.
Dashed line = an edge hidden behind a face.

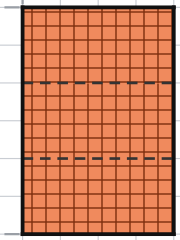
### FRONT VIEW

The steps show as steps.
This is the view that tells you the most,
so it goes in the front spot.



### RIGHT SIDE VIEW

From the right, the tallest step hides the other two.
Their top edges are behind it, so they are DASHED.



Views line up: TOP sits straight above FRONT, RIGHT sits straight beside it. Leave two squares between views. One cube = two squares.

# CVHS ENGINEERING

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## Example C: the T piece, hidden stem

Worked example. One cube = two squares.
Each colour is one side of the block. Find it in the picture.

Name _____

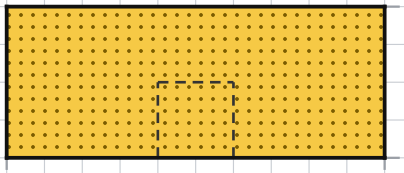
Date _____ Period _____

Colour = the side you are looking at

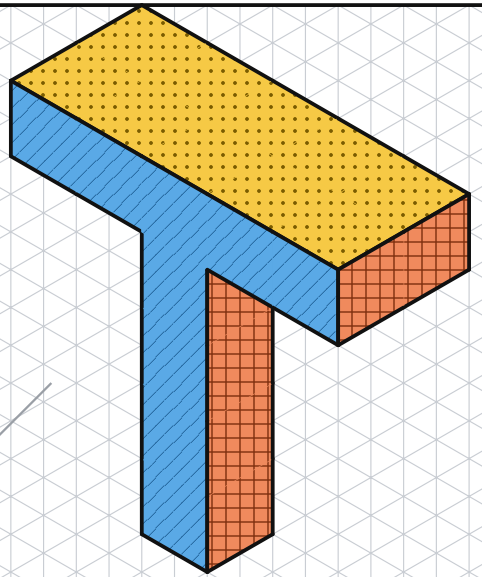
TOP FRONT RIGHT

### TOP VIEW

The stem is tucked under the bar,
so you can't see it from above.
Edges you can't see are DASHED.



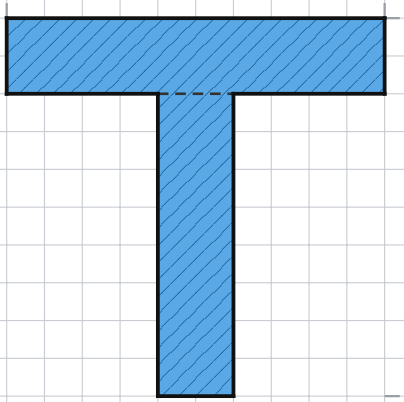
### ISOMETRIC



Look under the bar in the picture.
The stem only reaches 1 cube back.

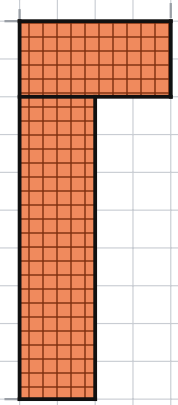
### FRONT VIEW

From the front it is a T.
The short dashed bit is the bottom edge of the bar,
hidden behind the stem.



### RIGHT SIDE VIEW

The bar is 2 deep, the stem is 1 deep.
The step shows as an L shape.



Views line up: TOP sits straight above FRONT, RIGHT sits straight beside it. Leave two squares between views. One cube = two squares.

# CVHS ENGINEERING

Iso + ortho sketch sheet | 5 mm grid

Third-angle layout. Sheet v0.3, Nov 2026.

## Problem 1: the heart

The picture is given. Draw the TOP, FRONT and RIGHT views.
Colour each view to match. Dashed = an edge you can't see.

Name _____

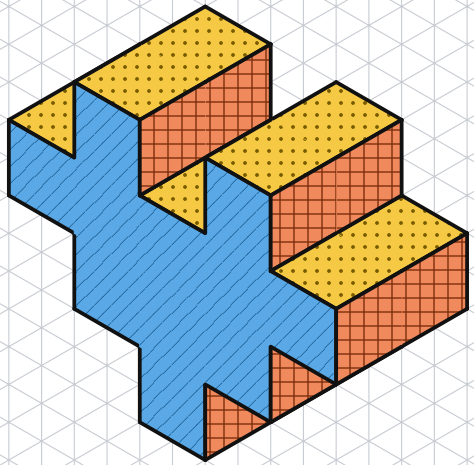
Date _____ Period _____

Colour = the side you are looking at

TOP FRONT RIGHT

### TOP VIEW

### ISOMETRIC



### FRONT VIEW

### RIGHT SIDE VIEW

| CHECK IT | Me | Teacher |
|---|---|---|
| Views line up | ☐ | ☐ |
| Cube counts match | ☐ | ☐ |
| Any hidden edge is dashed | ☐ | ☐ |

Views line up: TOP sits straight above FRONT, RIGHT sits straight beside it. Leave two squares between views. One cube = two squares.

# CVHS ENGINEERING

Iso + ortho sketch sheet | 5 mm grid

Third-angle layout. Sheet v0.3, Nov 2026.

## Problem 2: the number 6

The picture is given. Draw the TOP, FRONT and RIGHT views.
Colour each view to match. Dashed = an edge you can't see.

Name _____

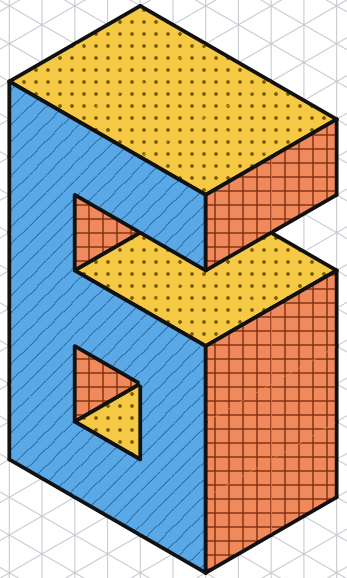
Date _____ Period _____

Colour = the side you are looking at

TOP FRONT RIGHT

### TOP VIEW

### ISOMETRIC



### FRONT VIEW

### RIGHT SIDE VIEW

| CHECK IT | Me | Teacher |
|---|---|---|
| Views line up | ☐ | ☐ |
| Cube counts match | ☐ | ☐ |
| Any hidden edge is dashed | ☐ | ☐ |

Views line up: TOP sits straight above FRONT, RIGHT sits straight beside it. Leave two squares between views. One cube = two squares.

# CVHS ENGINEERING

Iso + ortho sketch sheet | 5 mm grid

Third-angle layout. Sheet v0.3, Nov 2026.

## Problem 3: the number 7

The picture is given. Draw the TOP, FRONT and RIGHT views.
Colour each view to match. Dashed = an edge you can't see.

Name _____

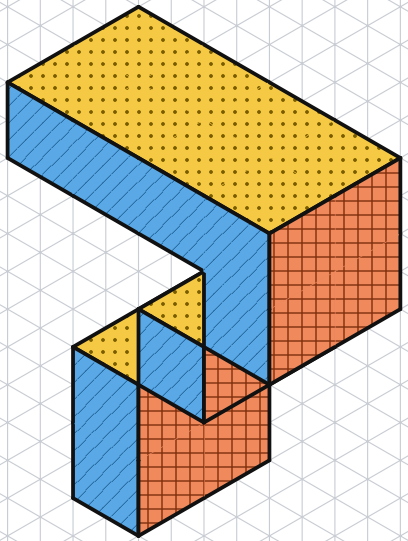
Date _____ Period _____

Colour = the side you are looking at

TOP FRONT RIGHT

### TOP VIEW

### ISOMETRIC



### FRONT VIEW

### RIGHT SIDE VIEW

| CHECK IT | Me | Teacher |
|---|---|---|
| Views line up | ☐ | ☐ |
| Cube counts match | ☐ | ☐ |
| Any hidden edge is dashed | ☐ | ☐ |

Views line up: TOP sits straight above FRONT, RIGHT sits straight beside it. Leave two squares between views. One cube = two squares.

# CVHS ENGINEERING

Iso + ortho sketch sheet | 5 mm grid

Third-angle layout. Sheet v0.3, Nov 2026.

## Problem 4: the ghost

The picture is given. Draw the TOP, FRONT and RIGHT views.
Colour each view to match. Dashed = an edge you can't see.

Name _____

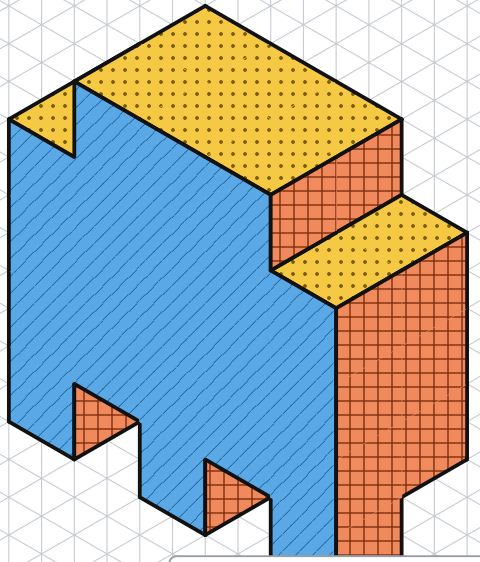
Date _____ Period _____

Colour = the side you are looking at

TOP FRONT RIGHT

### TOP VIEW

### ISOMETRIC



The bottom row (the skirt) is only 1 cube deep.
Everything else is 2 deep.

### FRONT VIEW

### RIGHT SIDE VIEW

| CHECK IT | Me | Teacher |
|---|---|---|
| Views line up | ☐ | ☐ |
| Cube counts match | ☐ | ☐ |
| Any hidden edge is dashed | ☐ | ☐ |

Views line up: TOP sits straight above FRONT, RIGHT sits straight beside it. Leave two squares between views. One cube = two squares.

# CVHS ENGINEERING

Iso + ortho sketch sheet | 5 mm grid

Third-angle layout. Sheet v0.3, Nov 2026.

## Problem 5: the mushroom

The three views are given. Draw the block on the iso grid.
Colour the faces: yellow top, blue front, orange right.

Name _____

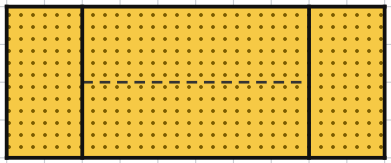
Date _____ Period _____

Colour = the side you are looking at

TOP FRONT RIGHT

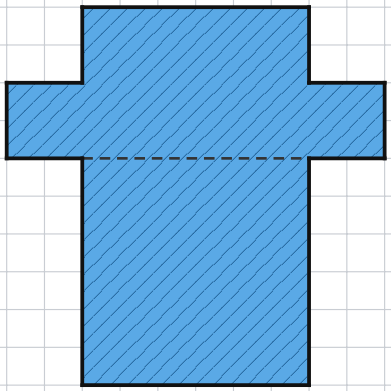
### TOP VIEW

Dashed lines: something is hidden under the cap.

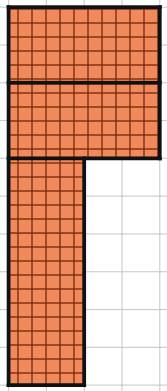


### ISOMETRIC

### FRONT VIEW



### RIGHT SIDE VIEW



| CHECK IT | Me | Teacher |
|---|---|---|
| Cube counts match the views | ☐ | ☐ |
| Set-back parts sit back | ☐ | ☐ |
| Each face the right colour | ☐ | ☐ |

Views line up: TOP sits straight above FRONT, RIGHT sits straight beside it. Leave two squares between views. One cube = two squares.

# CVHS ENGINEERING

Iso + ortho sketch sheet | 5 mm grid

Third-angle layout. Sheet v0.3, Nov 2026.

## Problem 6: the trophy

The three views are given. Draw the block on the iso grid.
Colour the faces: yellow top, blue front, orange right.

Name _____

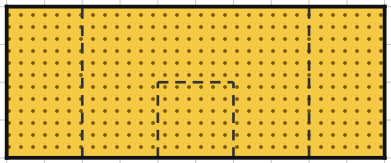
Date _____ Period _____

Colour = the side you are looking at

TOP FRONT RIGHT

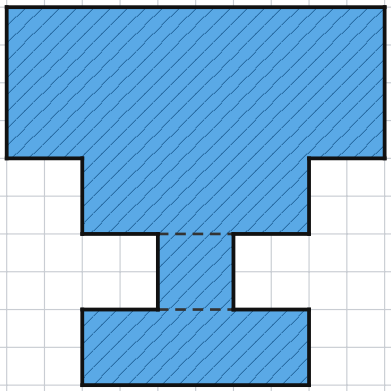
### TOP VIEW

One part is set back. The dashed lines show where.

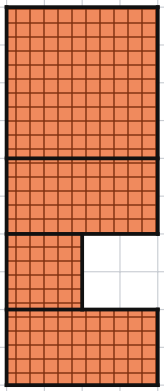


### ISOMETRIC

### FRONT VIEW



### RIGHT SIDE VIEW



| CHECK IT | Me | Teacher |
|---|---|---|
| Cube counts match the views | ☐ | ☐ |
| Set-back parts sit back | ☐ | ☐ |
| Each face the right colour | ☐ | ☐ |

Views line up: TOP sits straight above FRONT, RIGHT sits straight beside it. Leave two squares between views. One cube = two squares.

# CVHS ENGINEERING

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## Problem 7: the cactus

The three views are given. Draw the block on the iso grid.
Colour the faces: yellow top, blue front, orange right.

Name _____

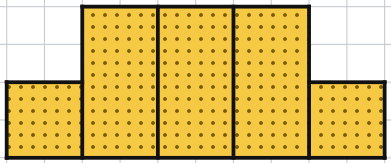
Date _____ Period _____

Colour = the side you are looking at

TOP FRONT RIGHT

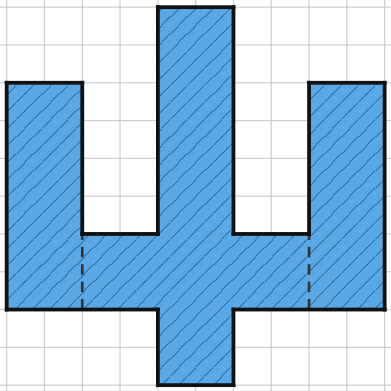
### TOP VIEW

The arms are thinner than the trunk.
Check the right view too.

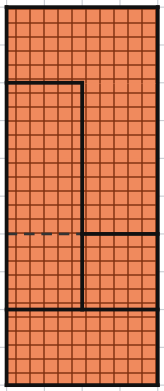


### ISOMETRIC

### FRONT VIEW



### RIGHT SIDE VIEW



| CHECK IT | Me | Teacher |
|---|---|---|
| Cube counts match the views | ☐ | ☐ |
| Set-back parts sit back | ☐ | ☐ |
| Each face the right colour | ☐ | ☐ |

Views line up: TOP sits straight above FRONT, RIGHT sits straight beside it. Leave two squares between views. One cube = two squares.